

Xiucheng “Andy” Zhang

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Education

New York University M.S. in Computer Science (Courant Institute) · GPA: 3.87/4.00 (through Spring 2026)	NYC, U.S. 09/2025 - 05/2027
Sun Yat-sen University B.S. in Artificial Intelligence · GPA: 3.80/4.00	Guangzhou, China 09/2021 - 06/2025

Research Interests

Natural Language Processing, LLM Agents, Human-AI Collaboration, Multi-Agent Systems, Agent Safety

Publications

* Equal contribution. † Corresponding author.

- [1] Zhuoning Xu*, **Xiucheng Zhang***, Hanjun Luo*, Yingbin Jin, Yinpeng Dong†, Hanan Salam. *MasDrift: Benchmarking Authorization Preservation Across Multi-Agent Architectures*. arXiv preprint 2026.
- [2] Ziqi Pan, **Xiucheng Zhang**, Zisu Li, Zhenhui Peng, Mingming Fan, Xiaojuan Ma†. *ACKnowledge: A Computational Framework for Human-Compatible Affordance-based Interaction Planning in Real-world Contexts*. CHI 2025.
- [3] **Xiucheng Zhang**, Yixin Zeng*, Qichang Li*, Guanyi Chen, Qian Yao Xu, Xiaozhu Hu, Zhenhui Peng†. *DesignWatch: Analyzing Users' Operations of Mobile Apps Based on Screen Recordings*. MobileHCI 2024 (Late-Breaking Work).
- [4] Yiwei Yuan, Zhiqing Wang, **Xiucheng Zhang**, Yichao Luo, Shuya Lin, Yang Bai, Zhenhui Peng†. *Exploring Community-Powered Conversational Agent for Health Knowledge Acquisition: A Case Study in Colorectal Cancer*. ICHEC 2025.
- [5] Yu Liang, Shilei Cao, **Xiucheng Zhang**, Juepeng Zheng†, Jianxi Huang, Haohuan Fu. *Low Saturation Confidence Distribution-based Test-Time Adaptation for Cross-Domain Remote Sensing Image Classification*. Int'l Journal of Applied Earth Observation and Geoinformation (JAG) 2025.

Research Experience

Research Assistant | New York University 07/2026 – Present

Disentangling Persona and Base-Model Effects in Long-Horizon Role Play Supervisor: Prof. Hanan Salam

Manuscript in preparation.

- Investigated whether base-model tendencies persist in persona-driven agents during long-horizon social interactions.
- Built book- and script-based simulations that replay character interactions from controlled narrative points against human-written continuations.
- Designed persona removal and model swap controls to distinguish persona effects from base-model effects.

Research Assistant | Tsinghua University 05/2026 – 07/2026

MasDrift: Authorization Preservation in Multi-Agent Architectures Supervisor: Prof. Yinpeng Dong

- Co-developed *MasDrift* and built 600 executable tasks to test whether user authorization survives delegation when permitted and reserved actions share the same tool environment.
- Evaluated agent architectures across 90,000 runs. Centralized hierarchies achieved 93.9–98.6% task completion but incurred 2.7–19.8% unauthorized-action rates, compared with 85.7–87.0% completion and 0.6–0.8% unauthorized actions in peer-based systems.

Research Assistant | Hong Kong University of Science and Technology

05/2024 – 09/2024

Context-aware Cognitive Architecture for LLM Task Planning

Supervisor: Prof. Xiaojuan Ma

- Co-developed *ACKnowledge*, an LLM planning agent that combines affordance graphs, visual grounding, and contextual reasoning for household tasks.
- Incorporated physical, personal, and social context into planning to support interpretable plans and user-guided revision. Evaluated the agent in simulation and real-world household scenarios against LLM-only baselines.

Research Assistant | Sun Yat-sen University

10/2023 – 09/2024

DesignWatch: Diagnosing Mobile-App Usability from Screen Recordings

Supervisor: Prof. Zhenhui Peng

- Proposed and led development of *DesignWatch*, a tool that extracts operation paths from mobile-app screen recordings via ResNet-18 key-frame selection and prompts GPT-4V to infer users' reasoning on UI transitions.
- Designed and optimized prompt structures for GPT-4V to infer action intent and usability issues; evaluated outputs against human UX-designer annotations.
- Collected data from 20 participants across 12 mobile apps and ran a mixed-method case study with 4 UX designers; *DesignWatch* reduced manual review time and surfaced usability issues unseen by baseline workflows.

Community-Powered Conversational Agent for Health Knowledge Acquisition

Supervisor: Prof. Zhenhui Peng

- Developed and deployed a RAG-based conversational agent on WeChat that integrates doctor-curated QA pairs with peer-led community posts using BERTopic, embedding-based retrieval, and expert-informed filtering.

Test-Time Adaptation for Cross-Domain Remote-Sensing Classification

Supervisor: Prof. Juepeng Zheng

- Led experimental evaluation and ablation of LSCD-TTA on AID, NWPU-RESISC45, and UC-Merced across six cross-domain tasks, achieving up to 5.22% accuracy improvement while maintaining real-time inference.

Professional Experience

Research Intern | Tencent

05/2026 – 08/2026

- Contributed to *MisPlan*, a diagnostic benchmark that derives five plan-content failure types from 125,275 real coding-agent sessions and evaluates 193 human-validated repository tasks under seven paired planning conditions.
- Supported large-scale evaluation across six models on a unified harness and three native coding agents, helping identify goal-scope misalignment as the most damaging planning failure, reducing issue-resolution rates by 25 percentage points relative to correct plans. Co-authored the resulting paper, currently under review at AAAI 2027.

Machine Learning Engineer Intern | Shanghai Xijin Information Technology Co., Ltd.

07/2023 – 09/2023

- Improved an LLM-powered customer-service QA agent by adapting prompts to a private knowledge base and building APIs for file conversion, semantic text segmentation, and downstream document processing.

Skills

Programming: Python, C, C++, SQL, Bash

ML Frameworks: PyTorch, Hugging Face Transformers

ML / NLP Methods: Retrieval-Augmented Generation, Embedding-Based Retrieval, Knowledge Graphs

Research Methods: Benchmark Design, Dataset Construction, Controlled Experiments, Large-Scale Agent Evaluation, Ablation Studies, Human Evaluation

Languages: Mandarin (native), English (professional)